

Solution

List of Matlab files:

Exercise 1	<code>getRandom.m</code>
Exercise 2	<code>getMoments.m</code>
Exercise 3	<code>Gauss.m</code>
Exercise 4	<code>getMean_median_mode.m</code>

Difficult points:

- bias of the variance estimator.

1 Random experiment

1. Simulate a random experiment:

Matlab file: `getRandom.m`:

- Let us consider the suitcases embarked on a plane.
 - Let us consider a set of X cases with a random number of cases between 100 and 150. The sample space is $[100; 150]$. It is a discrete random variable: $r = \text{randi}([100 \ 150], 1, 5)$.
 - Let us consider a set of 10 cases with random masses uniformly distributed between 20 kg and 30 kg. It is a continuous RV:

Uniform distribution $u = 20 + 10 * \text{rand}(1, 10)$
 Normal distribution $n = \text{normrnd}(25, 5, [1 \ 10])$
 Weibull distribution $w = 20 + 10 * \text{wblrnd}(0.5, 0.4, [1 \ 10])$

2 Moments of probability distribution

Theoretical inputs: Moments of a random variable

To exhibit some typical properties of a probability distribution, we generally compute their moments.

Mean

The expected value, or the mean value of the random value X is defined as:

$$\mathbb{E}\{X\} = \mu_X = \sum_{x_i \in \varepsilon} x_i P_X\{x_i\}, \quad \text{for a discrete random variable,} \quad (1)$$

$$\mathbb{E}\{X\} = \mu_X = \int_{\varepsilon} x f_X(x) dx, \quad \text{for a continuous random variable.} \quad (2)$$

- The operator $\mathbb{E}$ is a linear operator:

$$\mathbb{E}\{aX + bY\} = a \mathbb{E}\{X\} + b \mathbb{E}\{Y\}. \quad (3)$$

- Let consider a deterministic function g of the random variable X :

$$\begin{aligned} g: \mathbb{R} &\rightarrow \mathbb{R} \\ X &\mapsto g(x). \end{aligned}$$

The value of $g(X)$ is also a random variable, its expected value reads:

$$\mathbb{E}\{g(X)\} = \sum_{x_i \in \mathcal{E}} g(x_i) P_X\{x_i\}, \quad \text{for a discrete random variable,} \quad (4)$$

$$\mathbb{E}\{g(X)\} = \int_{\mathcal{E}} g(x) f_X(x) dx, \quad \text{for a continuous random variable.} \quad (5)$$

In general, $\mathbb{E}\{g(X)\} \neq g(\mathbb{E}\{X\})$.

Moments of order r

For general cases, the moment of order r of X is defined by:

$$\mathbb{E}\{X^r\} = \sum_{x_i \in \mathcal{E}} x_i^r P_X\{x_i\}, \quad \text{for a discrete random variable,} \quad (6)$$

$$\mathbb{E}\{X^r\} = \int_{\mathcal{E}} x^r f_X(x) dx, \quad \text{for a continuous random variable.} \quad (7)$$

The central moment of order r of X related to the mean value of the random variable μ_X previously defined, and not central around the origin of the coordinate system, is defined by:

$$\mathbb{E}\{(X - \mu_X)^r\} = \sum_{x_i \in \mathcal{E}} (x_i - \mu_X)^r P_X\{x_i\}, \quad \text{for a discrete random variable,} \quad (8)$$

$$\mathbb{E}\{(X - \mu_X)^r\} = \int_{\mathcal{E}} (x - \mu_X)^r f_X(x) dx, \quad \text{for a continuous random variable.} \quad (9)$$

The mean is the statistical moment of first order. Some other specific moments are detailed below.

Variance and Standard deviation

The variance is the central moment of order 2 of X . It represents the measure of the deviation of the random variable X :

$$\text{Var}\{X\} = \mathbb{E}\{(X - \mu_X)^2\} \quad (10)$$

It represents the moment of inertia of the probability density function about the mean value. The variance of a random variable is analogous to the moment of inertia of a weight about its centroid.

For a continuous random variable X ,

$$\begin{aligned} \text{Var}\{X\} &= \int_{\mathcal{E}} (x - \mu_X)^2 f_X(x) dx \\ &= \int_{\mathcal{E}} x^2 f_X(x) dx - \int_{\mathcal{E}} 2x\mu_X f_X(x) dx + \int_{\mathcal{E}} \mu_X^2 f_X(x) dx \\ &= \int_{\mathcal{E}} x^2 f_X(x) dx - 2\mu_X \int_{\mathcal{E}} x f_X(x) dx + \mu_X^2 \int_{\mathcal{E}} f_X(x) dx \\ &= \int_{\mathcal{E}} x^2 f_X(x) dx - 2\mu_X^2 + \mu_X^2 \end{aligned}$$

because $\mu_X = \int_{\mathcal{E}} x f_X(x) dx$ and $\int_{\mathcal{E}} f_X(x) dx = 1$. Therefore,

$$\text{Var}\{X\} = \mathbb{E}\{(X)^2\} - (\mathbb{E}\{X\})^2 \quad (11)$$

and the variance can be estimated as:

$$\text{Var}\{X\} = \sum_{x_i \in \varepsilon} x_i^2 P_X\{x_i\} - \mu_X^2, \quad \text{for a discrete random variable,} \quad (12)$$

$$\text{Var}\{X\} = \int_{\varepsilon} x^2 f_X(x) dx - \mu_X^2, \quad \text{for a continuous random variable.} \quad (13)$$

The coefficient of variation α_X of a random variable is given as:

$$\alpha_X = \frac{\sigma_X}{\mu_X}, \quad \text{if } \mu_X \neq 0. \quad (14)$$

The standard deviation σ_X of X is the positive square root of the variance σ_X^2 :

$$\sigma_X = \sqrt{\text{Var}(X)}. \quad (15)$$

Chebyshev's inequality From the variance σ_X^2 , the Chebyshev inequality provides a majoration of the probability that the distance between the values of a random variable X and its expected value μ_X is large:

$$\forall \alpha \in \mathbb{R}^+, \quad P(|X - \mu_X| \geq \alpha) \leq \frac{\sigma_X^2}{\alpha^2}. \quad (16)$$

Skewness

The moment of order 3, called skewness, is a measure for the symmetry of the distribution. It is defined as:

$$s_X = \mathbb{E} \left\{ \left(\frac{X - \mu_X}{\sigma_X} \right)^3 \right\}. \quad (17)$$

Kurtosis

The moments of order 4, called kurtosis, gives statements for the slope or the peak of the distribution:

$$\kappa_X = \mathbb{E} \left\{ \left(\frac{X - \mu_X}{\sigma_X} \right)^4 \right\}. \quad (18)$$

In some literature, the kurtosis is described by the excess κ_{X_e} , which is the difference between the kurtosis and the kurtosis of the normal distribution $\kappa_X^{normal} = 3$: $\kappa_{X_e} = \kappa_X - 3$. See Exercise ?? p.??

The information provided by all the moments can be provided by the characteristic function.

Concerning the continuous random variables, probability density functions are often classical mathematical functions which exhibit interesting and convenient numerical properties.

2.1 For discrete variables

For theoretical definitions, see Part 2 p.1.

2. Empirical mean

For this discrete random variable, from Equation (1), $\mathbb{E}\{X\} = \mu_X = \sum_{x_i \in \varepsilon} x_i P_i = \sum_{x_i \in \varepsilon} x_i * \frac{1}{N}$.

Matlab file: `getMoments.m`:

```
mean = sum(x)/numel(x) ;
```

3. Empirical variance and standard deviation

For this discrete random variable, from Equation (12), $\text{Var}\{X\} = \sum_{x_i \in \varepsilon} x_i^2 P_i - \mu_X^2 = \sum_{x_i \in \varepsilon} x_i^2 * \frac{1}{N} - \mu_X^2$.

Matlab file: `getMoments.m`:

```
variance = (1/numel(x) * sum(x.^2) - mean(x)^2 ) ;
```

```
\% VAR normalizes Y by N-1 if N>1, where N is the sample size. This is
\% an unbiased estimator of the variance of the population from which X is
\% drawn, as long as X consists of independent, identically distributed
\% samples. For N=1, Y is normalized by N.
```

```
if N>1
    var = var *N/(N-1) ;
end
```

The comments about the biased estimator should be included in Question 6.

Iterative computation of mean and variance

The mean may also be computed iteratively. For the first j samples, with $n < N$ and N being the maximum number of samples, the mean may be calculated as

$$\mu_{X_n} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^{n-1} x_i + \frac{1}{n} x_n \quad (19)$$

$$= \frac{n-1}{n} \frac{1}{n-1} \sum_{i=1}^{n-1} x_i + \frac{1}{n} x_n \quad (20)$$

$$= \frac{n-1}{n} \mu_{X_{n-1}} + \frac{1}{n} x_n \quad (21)$$

$$= \mu_{X_{n-1}} + \frac{(x_n - \mu_{X_{n-1}})}{n} \quad (22)$$

Similarly, the variance of the j samples may be computed in one single pass

$$\sigma_n^2 = \sigma_{n-1}^2 \frac{n-1}{n} + \frac{1}{n} (x_n - \mu_{X_{n-1}})^2 \quad (23)$$

4. Empirical skewness

For this discrete random variable, from Equation (17),

$$s_X = \mathbb{E} \left\{ \left(\frac{X - \mu_X}{\sigma_X} \right)^3 \right\} = \sum_i \left(\frac{X - \mu_X}{\sigma_X} \right)^3 P_X \{x_i\} = \sum_i \left(\frac{X - \mu_X}{\sigma_X} \right)^3 \frac{1}{N}.$$

Matlab file: `getMoments.m`:

```
skew = 0.0 ;
var = (1/numel(x) * sum(x.^2) - mean(x)^2 ) ;

for i=1:length(x)
    skew = skew + power(x(i)-mean(x),3)/(numel(x)*power(var,1.5));
end
```

N.B. This function uses the non-normalized variance.

5. Empirical kurtosis

For this discrete random variable, from Equation (18),

$$\kappa_X = \mathbb{E} \left\{ \left(\frac{X - \mu_X}{\sigma_X} \right)^4 \right\} = \sum_i \left(\frac{X - \mu_X}{\sigma_X} \right)^4 P_X \{x_i\} = \sum_i \left(\frac{X - \mu_X}{\sigma_X} \right)^4 \frac{1}{N}.$$

Matlab file: `getMoments.m`:

```
kurt = 0.0 ;
variance = (1/numel(x) * sum(x.^2) - mean(x)^2 ) ;

for i=1:length(x)
    kurt = kurt + power(x(i)-mean(x),4)/(numel(x)*power(variance,2.0));
end
```

N.B. This function uses the non-normalized variance.

6. Comparison between the implemented functions and the functions provided by Matlab

Matlab file: `getMoments.m`: Compare the two lines of the output matrix. The variances computed by the ourselves implemented function `getMoments.m` and using `Matlab` should be different. `Matlab` normalizes the variance by $N - 1$ if $N > 1$ to obtain an estimator whose bias vanishes. The quality of an estimator $\hat{\theta}$ of the quantity θ is provided by its bias:

$$\text{Bias} \left\{ \hat{\theta} \right\} = \mathbb{E} \left\{ \hat{\theta} \right\} - \theta. \quad (24)$$

Variance estimator from Eq.(12):

$$\begin{aligned} \widehat{\text{Var}} \{X\} &= \sum_{i=1}^N \frac{1}{N} x_i^2 - [\mathbb{E} \{X\}]^2 \\ \text{Bias} \left(\widehat{\text{Var}} \{X\} \right) &= \mathbb{E} \left\{ \widehat{\text{Var}} \{X\} \right\} - \text{Var} \{X\} \\ &= \mathbb{E} \left\{ \frac{1}{n} \sum_{i=1}^n x_i^2 - \mathbb{E} \{X\}^2 \right\} - \text{Var} \{X\} \end{aligned} \quad (25)$$

As we know that $\text{Var} \{X\} = \mathbb{E} \{X^2\} - \mathbb{E} \{X\}^2$ from Eq.(11), thus

$$\begin{aligned} \mathbb{E} \left\{ \mathbb{E} \{X\}^2 \right\} &= \text{Var} \{ \mathbb{E} \{X\} \} + \mathbb{E} \{ \mathbb{E} \{X\} \}^2 \\ &= \frac{1}{N} \text{Var} \{X\} + \left[\sum_{i=1}^N \mathbb{E} \{X\} \frac{1}{N} \right]^2 \\ &= \frac{1}{N} \text{Var} \{X\} + [\mathbb{E} \{X\}]^2 \end{aligned}$$

$$\begin{aligned}\mathbb{E} \left\{ \frac{1}{N} \sum_{i=1}^N x_i^2 \right\} &= \frac{1}{N} \mathbb{E} \left\{ \sum_{i=1}^N x_i^2 \right\} = \frac{1}{N} \sum_{i=1}^N \mathbb{E} \{ x_i^2 \} \\ &= \frac{1}{N} N \mathbb{E} \{ X^2 \} = \mathbb{E} \{ X^2 \} \\ &= \mathbb{E} \{ X \}^2 + \text{Var} \{ X \}.\end{aligned}$$

$$\begin{aligned}\mathbb{E} \left\{ \widehat{\text{Var}} \{ X \} \right\} &= \mathbb{E} \left\{ \frac{1}{N} \sum_{i=1}^N x_i^2 - \mu_X^2 \right\} \\ &= [\mathbb{E} \{ X \}]^2 + \text{Var} \{ X \} - [\mathbb{E} \{ X \}]^2 - \frac{1}{N} \text{Var} \{ X \} \\ &= \frac{N-1}{N} \text{Var} \{ X \}\end{aligned}$$

With the normalization, we obtain an estimator without bias even when $N > 1$.

$$\text{Bias} \left(\frac{n}{n-1} \widehat{\text{Var}} \{ X \} \right) = \mathbb{E} \left\{ \frac{n}{n-1} \widehat{\text{Var}} \{ X \} \right\} - \text{Var} \{ X \} = \frac{n}{n-1} \frac{n-1}{n} \text{Var} \{ X \} - \text{Var} \{ X \} = 0$$

7. Sets of data

- Same mean, same skewness but difference variances:

$$X_1 = [0; 1; 2; 3; 4; 5], \quad X_2 = [-5; 1; 2; 3; 4; 10]$$

- Same mean, but different variances and different skewness:

$$X_1 = [0; 1; 2; 3; 4; 5], \quad X_2 = [0; 1; 2; 2; 5; 5]$$

- Positive skewness:

$$X = [0; 1; 0; 1; 0; 5]$$

- Negative skewness:

$$X = [0; 4; 5; 4; 5; 5]$$

- Same mean, but different kurtosis:

$$X_1 = [0; 2; 4], \quad X_2 = [0; 2; 2; 2; 4]$$

- Positive kurtosis:

$$X = [-5; 1; 2; 3; 4; 10]$$

- Negative kurtosis: here, we can only get kurtosis ≥ 0

8. Which of these moments exhibit linear properties?

- the expected value is a linear operator: $\mathbb{E} \{ aX + bY \} = a\mathbb{E} \{ X \} + b\mathbb{E} \{ Y \}$,
- the variance is a non-linear operator,
- the skewness is a non-linear operator,
- the kurtosis is a non-linear operator.

2.2 For some classical continuous distributions

Uniform distribution

9. For a uniform distribution law, establish the analytical expressions of the expected value, the variance, the skewness and the kurtosis.

A random variable X with uniform distribution between the limits a and b is described by the probability density function f_X^{Uniform} :

$$f_X^{\text{Uniform}}(x) = \begin{cases} \frac{1}{b-a} & \forall x \in [a, b], \\ 0 & \forall x \notin [a, b], \end{cases}$$

or the cumulative distribution function F_X^{Uniform} :

$$F_X^{\text{Uniform}}(x) = \begin{cases} \int_a^x \frac{1}{b-a} dt = \left[\frac{x}{b-a} \right]_a^x = \frac{x-a}{b-a} & \forall x \in [a, b], \\ 0 & \forall x \notin [a, b]. \end{cases}$$

2.2.1 Expected value

$$\mu^{\text{Uniform}} = \int_a^b x f_X^{\text{Uniform}}(x) dx = \int_a^b \frac{x}{b-a} dx = \left[\frac{x^2}{2(b-a)} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2} \quad (26)$$

2.2.2 Variance

$$\begin{aligned} \text{Var}\{X^{\text{Uniform}}\} &= \int_a^b (x - \mu^{\text{Uniform}})^2 f_X(x) dx \\ &= \int_a^b \left(x - \frac{a+b}{2} \right)^2 \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \left[\frac{1}{3} \left(x - \frac{a+b}{2} \right)^3 \right]_a^b \\ &= \frac{1}{3(b-a)} \left(\left(b - \frac{a+b}{2} \right)^3 - \left(a - \frac{a+b}{2} \right)^3 \right) \\ &= \frac{1}{3(b-a)} \left(\left(\frac{b-a}{2} \right)^3 - \left(\frac{a-b}{2} \right)^3 \right) \\ \text{Var}\{X^{\text{Uniform}}\} &= \frac{(a-b)^2}{12} \end{aligned} \quad (27)$$

2.2.3 Skewness

$$\begin{aligned}
 s_X^{Uniform} &= \mathbb{E} \left\{ \left(\frac{X - \mu_X}{\sigma_X} \right)^3 \right\} \\
 &= \int_a^b \left(\frac{x - \mu^{Uniform}}{\sigma_X^{Uniform}} \right)^3 f_X(x) dx \\
 &= \int_a^b \left(x - \frac{a+b}{2} \right)^3 \left(\frac{2\sqrt{3}}{b-a} \right)^3 \frac{1}{b-a} dx \\
 &= \left(\frac{2\sqrt{3}}{b-a} \right)^3 \frac{1}{b-a} \left[\frac{1}{4} \left(x - \frac{a+b}{2} \right)^4 \right]_a^b \\
 &= \left(\frac{2\sqrt{3}}{b-a} \right)^3 \frac{1}{b-a} \frac{1}{4} \left(\left(\frac{-a+b}{2} \right)^4 - \left(\frac{a-b}{2} \right)^4 \right) \\
 s_X^{Uniform} &= 0
 \end{aligned} \tag{28}$$

2.2.4 Kurtosis

$$\begin{aligned}
 \kappa_X^{Uniform} &= \mathbb{E} \left\{ \left(\frac{X - \mu_X}{\sigma_X} \right)^4 \right\} \\
 &= \int_a^b \left(\frac{x - \mu^{Uniform}}{\sigma_X^{Uniform}} \right)^4 f_X(x) dx \\
 &= \int_a^b \left(x - \frac{a+b}{2} \right)^4 \left(\frac{2\sqrt{3}}{b-a} \right)^4 \frac{1}{b-a} dx \\
 &= \left(\frac{2\sqrt{3}}{b-a} \right)^4 \frac{1}{b-a} \left[\frac{1}{5} \left(x - \frac{a+b}{2} \right)^5 \right]_a^b \\
 &= \left(\frac{2\sqrt{3}}{b-a} \right)^4 \frac{1}{b-a} \frac{1}{5} \left(\left(\frac{-a+b}{2} \right)^5 - \left(\frac{a-b}{2} \right)^5 \right) \\
 \kappa_X^{Uniform} &= \frac{2^4 \cdot 3^2 \cdot 2}{5 \cdot 2^5} = \frac{9}{5} = 1.8
 \end{aligned} \tag{29}$$

Exponential distribution

The exponential distribution of the random variable X with parameter $\lambda \in \mathbb{R}^+$ is defined by the probability density function $f_X^{\text{Exponential}}$:

$$f_X^{\text{Exponential}}(x) = \begin{cases} \lambda e^{-\lambda x} & \forall x \in [0, +\infty[, \\ 0 & \forall x \in]-\infty, 0[, \end{cases} \tag{30}$$

or by its cumulative distribution function $F_X^{\text{Exponential}}$:

$$F_X^{\text{Exponential}}(x) = \begin{cases} \int_0^x \lambda e^{-\lambda t} dt = \frac{-\lambda}{\lambda} [e^{-\lambda t}]_0^x = 1 - e^{-\lambda x} & \forall x \in [0, +\infty[, \\ \int_{-\infty}^x 0 dt = 0 & \forall x \in]-\infty, 0[. \end{cases} \tag{31}$$

We can check that:

$$\int_0^{+\infty} \lambda e^{-\lambda x} dx = \left[\frac{\lambda e^{-\lambda x}}{-\lambda} \right]_0^{+\infty} = 0 + \frac{\lambda}{\lambda} = 1 \tag{32}$$

2.2.5 Expected value

The mean value reads:

$$\begin{aligned}
 \mathbb{E}\{X\} &= \int_{\mathbb{R}} x f_X(x) dx \\
 &= \int_{-\infty}^0 x \cdot 0 dx + \int_0^{+\infty} x \lambda e^{-\lambda x} dx \\
 &= [A]_{-\infty}^0 + [-x e^{-\lambda x}]_0^{+\infty} + \int_0^{+\infty} 1 \cdot e^{-\lambda x} dx \\
 &= A - A + 0 - 0 + \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^{+\infty} \\
 &= 0 - \frac{1}{-\lambda} \\
 \mathbb{E}\{X\} &= \frac{1}{\lambda}
 \end{aligned} \tag{33}$$

2.2.6 Variance

$$\begin{aligned}
 \text{Var}\{X\} &= \int_a^b (x - \mu)^2 f_X(x) dx \\
 &\text{or} \\
 \text{Var}\{X\} &= \mathbb{E}\{X^2\} - (\mathbb{E}\{X\})^2 \\
 &= \int_{-\infty}^0 x^2 \cdot 0 dx + \int_0^{+\infty} x^2 \lambda e^{-\lambda x} dx - \frac{1}{\lambda^2} \\
 &= 0 - 0 + [-x^2 e^{-\lambda x}]_0^{\infty} + 2 \int_0^{\infty} x e^{-\lambda x} dx - \frac{1}{\lambda^2} \\
 &= 0 - 0 + 0 - 0 + 2 \left[-x \frac{e^{-\lambda x}}{\lambda} \right]_0^{\infty} + \frac{2}{\lambda} \int_0^{\infty} e^{-\lambda x} dx - \frac{1}{\lambda^2} \\
 &= \frac{2}{\lambda^2} - \frac{1}{\lambda} = \frac{1}{\lambda^2}
 \end{aligned} \tag{34}$$

2.2.7 Skewness

$$\begin{aligned}
 \text{Skew}\{X\} &= \mathbb{E} \left\{ \left(\frac{x - \mu}{\sigma} \right)^3 \right\} = \frac{\mathbb{E}\{x^3\} - 3\mu\sigma^2 - \mu^3}{\sigma^3} \\
 &= \frac{\frac{6}{\lambda^3} - 3 \frac{1}{\lambda} \frac{1}{\lambda^2} - \frac{1}{\lambda^3}}{\frac{1}{\lambda^3}} \\
 &= 2
 \end{aligned} \tag{35}$$

2.2.8 Kurtosis

$$\begin{aligned}
 \text{Kurt}\{X\} = \kappa\{X\} &= \mathbb{E} \left\{ \left(\frac{x - \mu}{\sigma} \right)^4 \right\} = \frac{\mathbb{E}\{x^4\} - 4\mu\mathbb{E}\{x^3\} + 6\mu^2\mathbb{E}\{x^2\} - 4\mu^3\mathbb{E}\{x\} + \mu^4}{\sigma^4} \\
 &= \frac{\mathbb{E}\{x^4\} - 4\mu\mathbb{E}\{x^3\} + 6\mu^2\mathbb{E}\{x^2\} - 3\mu^4}{\sigma^4} \\
 &= \frac{\mathbb{E}\{x^4\} - 4\mu\mathbb{E}\{x^3\} + 6\mu^2\sigma^2 + 3\mu^4}{\sigma^4} \\
 &= 9
 \end{aligned} \tag{36}$$

Sometimes the kurtosis is written as the “excess kurtosis”, which is the difference of the kurtosis of the distribution with respect to the normal distribution kurtosis (i.e. $\kappa\{X\} = \kappa\{X\} - 3$) and hence the “excess kurtosis” of an exponential is 6.

Gaussian distribution

See printed document.

3 A Gaussian continuous distribution

10. Investigate a Gaussian continuous distribution:

- Plot Gaussian probability density function and Gaussian cumulative distribution function for different mean values,

Matlab file: Gauss.m:

```
x = [mu-20*sigma:0.1*sigma:mu+20*sigma];
pdf_ref = 1/(sqrt(2*pi*sigma^2))*exp(-power(x-mu,2)/(2*sigma^2));
cdf_ref = cdf('Normal',x,mu,sigma);

% define three different values of the means
mu2 = 0.5*mu ;
mu3 = 2*mu ;
mu4 = 10*mu ;

% compute the 3 relative probability density functions
pdf_mu2 = 1/(sqrt(2*pi*sigma^2))*exp(-power(x-mu2,2)/(2*sigma^2));
pdf_mu3 = 1/(sqrt(2*pi*sigma^2))*exp(-power(x-mu3,2)/(2*sigma^2));
pdf_mu4 = 1/(sqrt(2*pi*sigma^2))*exp(-power(x-mu4,2)/(2*sigma^2));

% compute the 3 relative cumulative distribution functions
cdf_mu2 = cdf('Normal',x,mu2,sigma);
cdf_mu3 = cdf('Normal',x,mu3,sigma);
cdf_mu4 = cdf('Normal',x,mu4,sigma);

% plot the different pdfs
h3=figure(3);
plot(x,pdf_ref,'k','Linewidth',2)
hold on
plot(x,pdf_mu2,'r','Linewidth',2);
plot(x,pdf_mu3,'b','Linewidth',2);
plot(x,pdf_mu4,'g','Linewidth',2);
set(gca, 'FontSize', 15)
set(gca, 'linewidth', 2)
xlabel('Sample space','Interpreter','Latex','FontSize',20)
xlim ([mu-20*sigma mu+20*sigma])
ylabel('Probability density function','Interpreter','Latex','FontSize',20)

leg3{1}=strcat('\mu = ',num2str(mu));
leg3{2}=strcat('\mu = ',num2str(0.5*mu));
leg3{3}=strcat('\mu = ',num2str(2*mu));
leg3{4}=strcat('\mu = ',num2str(10*mu));

legend(leg3)
hold off
savefig(h3,'Gaussian_sigmas_pdf.fig');
%close(h3)

h4=figure(4);
plot(x,cdf_ref,'k','Linewidth',2);
hold on
plot(x,cdf_mu2,'r','Linewidth',2);
plot(x,cdf_mu3,'b','Linewidth',2);
plot(x,cdf_mu4,'g','Linewidth',2);
```

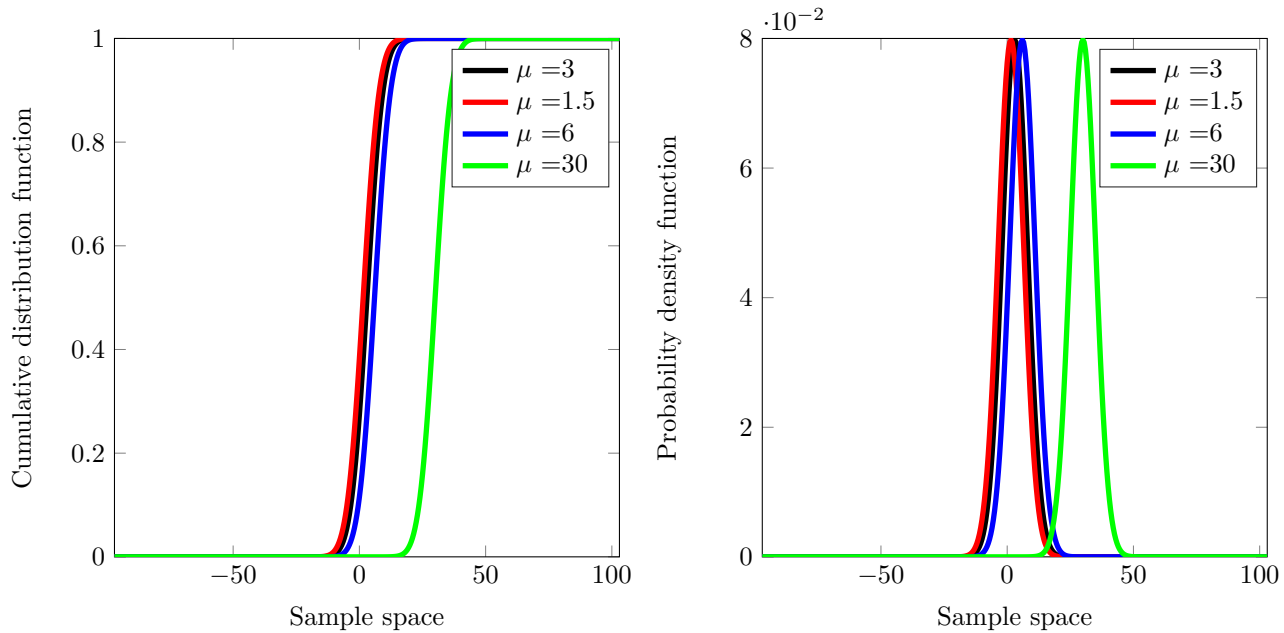


Figure 1: Probability density function and cumulative distribution function of Gaussian distributions with different mean values

```
leg4{1}=strcat('\mu = ',num2str(mu));
leg4{2}=strcat('\mu = ',num2str(0.5*mu));
leg4{3}=strcat('\mu = ',num2str(2*mu));
leg4{4}=strcat('\mu = ',num2str(10*mu));

legend(leg4)

set(gca, 'FontSize', 15)
set(gca, 'linewidth', 2)
xlabel('Sample space','Interpreter','Latex','FontSize',20)
xlim([mu-20*sigma mu+20*sigma])
ylabel('Cumulative distribution function','Interpreter','Latex','FontSize',20)
hold on
savefig(h4,'Gaussian_sigmas_cdf.fig');
```

- Plot Gaussian probability density function and Gaussian cumulative distribution function for different values of the standard deviation of the random variable,

Matlab file: Gauss.m:

```
% define three different values of the standard deviations
sigma2 = 0.5*sigma;
sigma3 = 2*sigma;
sigma4 = 10*sigma;

% compute the 3 relative probability density functions
pdf_sig2 = 1/(sqrt(2*pi*sigma2^2))*exp(-power(x-mu,2)/(2*sigma2^2));
pdf_sig3 = 1/(sqrt(2*pi*sigma3^2))*exp(-power(x-mu,2)/(2*sigma3^2));
pdf_sig4 = 1/(sqrt(2*pi*sigma4^2))*exp(-power(x-mu,2)/(2*sigma4^2));

% compute the 3 relative cumulative distribution functions
```

```

cdf_sig2 = cdf('Normal',x,mu,sigma2);
cdf_sig3 = cdf('Normal',x,mu,sigma3);
cdf_sig4 = cdf('Normal',x,mu,sigma4);

% plot the diffents pdfs
h1=figure(1);
plot(x,pdf_ref,'k','Linewidth',2)
hold on

% yy = pdf('Normal',x,mu,sigma);
% plot(x,yy,'r','Linewidth',2);
hold on
plot(x,pdf_sig2,'r','Linewidth',2);
plot(x,pdf_sig3,'b','Linewidth',2);
plot(x,pdf_sig4,'g','Linewidth',2);
set(gca, 'FontSize', 15)
set(gca, 'linewidth', 2)
xlabel( 'Sample space','Interpreter','Latex','FontSize',20)
xlim ([mu-20*sigma mu+20*sigma])
ylabel('Probability density function','Interpreter','Latex','FontSize',20)

leg1{1}=strcat('\sigma = ',num2str(sigma));
leg1{2}=strcat('\sigma = ',num2str(0.5*sigma));
leg1{3}=strcat('\sigma = ',num2str(2*sigma));
leg1{4}=strcat('\sigma = ',num2str(10*sigma));

legend(leg1)
hold off
savefig(h1,'Gaussian_sigmas_pdf.fig');
%close(h1)

% plot the different cdfs
h2=figure(2);
plot(x,cdf_ref,'k','Linewidth',2);
hold on
plot(x,cdf_sig2,'r','Linewidth',2);
plot(x,cdf_sig3,'b','Linewidth',2);
plot(x,cdf_sig4,'g','Linewidth',2);

leg2{1}=strcat('\sigma = ', num2str(sigma));
leg2{2}=strcat('\sigma = ',num2str(0.5*sigma));
leg2{3}=strcat('\sigma = ',num2str(2*sigma));
leg2{4}=strcat('\sigma = ',num2str(10*sigma));

legend(leg2)

set(gca, 'FontSize', 15)
set(gca, 'linewidth', 2)
xlabel( 'Sample space','Interpreter','Latex','FontSize',20)
xlim ([mu-20*sigma mu+20*sigma])
ylabel('Cumulative distribution function','Interpreter','Latex','FontSize',20)
hold on

```

- Which are the skewness and the kurtosis of these different distributions?
The skewness is equal to 0, and the kurtosis is equal to 3 (the excess kurtosis is equal to 0).
- From the probability density function or from the cumulative distribution function, determine the probability that x belongs to $[a; b]$.

Matlab file: Gauss.m:

From the PDF:

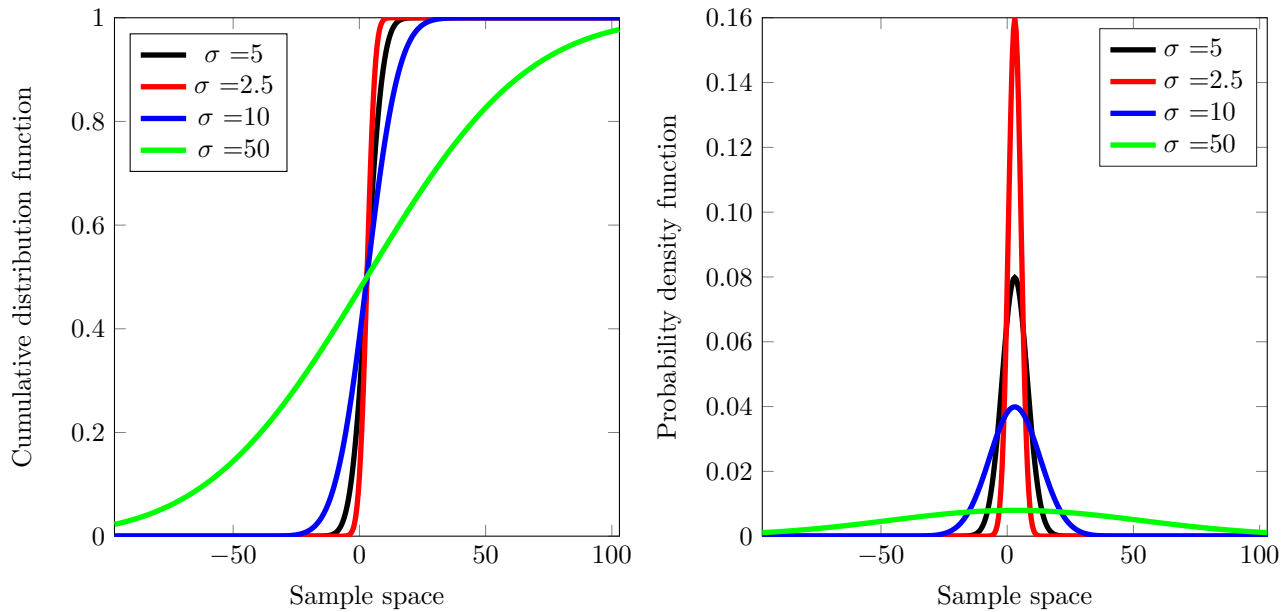


Figure 2: Probability density function and cumulative distribution function of Gaussian distributions with different standard deviation values

```
fun = @(x)(1/(sigma*sqrt(2*pi))*exp(-1/2*((x-mu)/sigma).^2));
P_ab_pdf=integral(fun,a,b);

From the CDF:
x = [-1500:0.05:1500];
z = cdf('Normal',x,mu,sigma);

% Probability that x belongs to (a,b)
a = -20;
b = 20;

% find indices relative to a and b
i_a = int64(1+(a-x(1))/(x(2)-x(1))) ;
i_b = int64(1+(b-x(1))/(x(2)-x(1))) ;

P_ab_cdf = cdf_ref(i_b) - cdf_ref(i_a);
```

The areas under the curve within one, two, and three standard deviations are about 68 %, 95.5 % and 99.7% of the total area, respectively.

Any linear function of normally distributed random variables is also normally distributed. A non-linear function of normally distributed random variables may or may not be normal.

4 Mean, median and mode

11. Compute the mean, the median and the mode for the discrete set of data: {2; 2; 2; 15; 36; 10}.

Matlab file: getMean_median_mode.m:

```
x = [2 2 2 15 36 10];
```

mean:

```
mean(x) = 11.1667
```

median:

```
median(x) = 6
```

The median value is the mean of the middle two numbers in sorted order.

mode:

```
mode(x) = 2
```

Mode, mean and median are different tools to provide a unique number to describe a distribution.

12. What are the median and the mode for a Gaussian distribution with mean equal to 3?

```
x = normrnd(3,15,25000000,1);
```

```
A = mean(x);
```

```
B = median(x);
```

```
C = mode(x);
```

We can observe that the estimation of the mean and the median are far from 3 when the sample has a small size, and that the estimation of the mode is not consistent with what would be expected.

```
C1 = round(C,3,'decimal');
```

The estimation of the mode shows that 3.00 is not the most occurent value. Arrounding with no decimals, 3 is the most occurent value.

We could compute mean, mode and median for other distributions.